A Survey on Synthetic Data Generation for Diabetes Applications

K.Sathyaseelan[1], Dr.S.Sarathambekai[2], Dr.R.Senthilkumar[3], Dr.S.Gokulraj[4]
Assistant Professor, Department of Artificial Intelligence and Data Science, Kalaignarkarunanidhi Institute of Technology, Coimbatore, India [1,]
Associate Professor, Department of Information Technology, PSG College of Technology, Coimbatore, India [2],
Associate Professor, Department of Computer Science and Engineering, Shree Venkateshwara Hi Tech Engineering College, Gobichettipalayam. India [3],
Associate Professor, Department of Computer Science and Engineering(CS), :Dr.N.G.P.Institute of Technology, Coimbatore, India [4].
Email: kitsathyaseelan@gmail.com

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Abstract - **Synthetic data is increasingly used in diabetes research for solving limited data availability, class imbalance, and patient privacy concerns. This review examines methods for generating synthetic data. It is categorized into four main groups like statistical methods, machine learning approaches, deep learning models and hybrid or emerging frameworks. The use of synthetic data is described across several diabetes related applications. Including electronic health records, continuous glucose monitoring time-series data, medical imaging, decision support systems and data augmentation for predictive modeling. Common real and synthetic datasets were reviewed, highlighting their strengths, weakness and relevance for diabetes studies. The evaluation strategies are reviewed, particularly on data fidelity, model performance and privacy preservation. Several challenges still need to be addressed. This include verifying the reliability of synthetic data and reducing the bias within algorithms. In addition, compliance with regulations like HIPAA and GDPR must be maintained while ensuring the solutions remain practical for real world use. The review highlights several directions for future research. These include advancing generative models and developing personalized with cross domain synthetic data. It also highlights the need for clear ethical and regulatory guidelines to ensure responsible use in clinical settings.**

Keywords - Synthetic data generation, Diabetes dataset, Machine and Deep Learning techniques, Class Imbalance.

I. INTRODUCTION

Diabetes mellitus (DM) is a long term chronic metabolic condition. Its prevalence is increasing worldwide. As a result, DM poses a major challenge to healthcare systems across the globe. According to the International Diabetes Federation, the number of adults living with diabetes continuous to rise. It is projected to reach 643 million by 2030, 783 million by 2045, and 853 million by 2050 [1]. Effective prediction and early diagnosis of diabetes are important. Ongoing management of diabetes also plays a key role. Together, these approaches help reduce complications and improve patient outcomes. In recent years, data centric methods have become more widely used in diabetes management. Techniques based on machine learning (ML) and artificial intelligence (AI) have shown considerable potential. These approaches have been particularly effective in evaluating risk, predicting blood glucose trends and the development of individual treatment strategies[2].

Even with recent advancements, creating strong ML models for diabetes remains challenging. The problems like limited data, high class imbalance and the sensitive patient information[3], [4]. Most clinical datasets are not used as direct inputs to ML or DL models. However, they often have the problems like small sample sizes, missing values and class imbalance. Some medical datasets follow strict privacy regulatory frameworks like HIPAA and GDPR which limits data sharing and reuse[5]. These issues make it harder to train generalized models and slow down innovation.

Generating synthetic data is increasingly recognized as an effective way to address these limitations[6]. It works by producing artificial but realistic data that preserves anonymity. It also closely matches the statistical properties of real patient records. Synthetic data help to reduce issues including data scarcity, privacy, and imbalances[7]. Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are commonly used for synthetic data generation. Specialized data augmentation techniques have made it possible to produce high quality synthetic medical data. These methods help develop more accurate and fair ML models. They also allow privacy preserving data sharing.

The broader field of synthetic data generation has attracted a lot of attention. Their specific applications in diabetes-related research are emerging and scattered[8]. Research on synthetic data is increasing. Many recent studies are exploring its use specifically in the diabetes area. However, these efforts are fragmented. They use a variety of methodologies and often lack a comparative or systematic approach. This survey provides a comprehensive overview of synthetic data generation approaches for diabetes applications. It can serve as a useful resource for diabetic based researchers.

The purpose of this survey was to address this research gap. It provides a structured and detailed review of synthetic data generation techniques used in diabetes domain. This survey examines key methods, datasets, evaluation metrics, use cases, and challenges. Finally future research directions are highlighted to support guide the research community.

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The primary contributions of this study include:

- To provide a comprehensive taxonomy of synthetic data-generation methods applicable to diabetes research.
- Review and categorization of existing studies by application, generation, and evaluation strategies.
- The practical challenges and ethical considerations associated with synthetic diabetes data are discussed.
- Identify open problems and future directions to foster innovation in this area of study.

The rest of this paper is structured as follows. Section 2 offers background information on diabetes and the role of data in ML-based healthcare. Section 3 introduces synthetic data-generation techniques, including deep generative models and classical approaches. Section 4 presents detailed applications of synthetic data for diabetes. Section 5 describes the various datasets and benchmarks for diabetes. Section 6 discusses the evaluation metrics and Section 7 highlights the current challenges and future research directions. Section 9 provides the conclusion of the paper.

II. BACKGROUND

A. Diabetes as a Global Health Challenge

DM represents a notable and most dangerous key international health concerns of the 21st century. Diabetes is defined by persistent high blood sugar levels caused by problems with insulin secretion, insulin function, or both. Diabetes can manifest in several manners. The most prevalent conditions are type1 diabetes (T1D)[9], an autoimmune condition that leads to absolute insulin deficiency, type 2 diabetes (T2D)[10], which is associated with insulin resistance and relative insulin deficiency. Gestational diabetes develops during pregnancy and poses health risks to both the mother and the unborn baby [11].

As of 2021, the International Diabetes Federation (IDF) has estimated that over 537 million adults are currently living with diabetes, with estimates projecting this number to rise to 783 million by 2045[1]. Diabetes can lead to many problems. It includes heart disease, nerve damage, kidney issues, eye problems, limb loss[12]. The socioeconomic impact was significant. It included direct medical costs, reduced productivity and increased mortality rates. Together these problems highlight the importance of timely diagnosis. They also underscore the need for timely diagnosis, effective monitoring and personalized treatment strategies.

B. The Imporatnce of data in diabetes detection and Management

Healthcare has become more digital than ever. This change has made it easier to use AI and ML in healthcare. Doctors now use data from many sources on diabetes care. These include health records, glucose monitors, wearables, genetic tests, insulin pumps, sensors, and lifestyle tracking [13]. Most of the data were obtained from heterogeneous sources. ML models are used in diabetes care in several ways. They can predict when the disease may develop. They can also identify complications, blood sugar patterns. Suggestions also made for personalized treatment[14]. On the other hand, performance of these models depends strongly on the available training data. Their ability to generalize is influenced by the amount, quality and diversity of the data. Effective AI applications for diabetes rely on access to large and well-labelled datasets. These datasets should be representative and include diverse demographic groups, comorbidities and clinical scenarios.

C. Challenges in real-world diabetes data

Although more health data are currently available. Still researchers and doctors face many challenges. Using real-world data for diabetes research remains complex and difficult.

- Data Scarcity and Heterogeneity: Many diabetes datasets are small. It often comes from only one hospital or clinic. In addition, they are difficult to share due to privacy rules and data ownership issues[15].
- Class Imbalance: Most datasets contain more records for certain groups than for others. For example, more healthy patients than for those with disease. This imbalance can may lead to biased models that work better for the majority groups[16].
- Missing and Incomplete Data: Data can sometimes be missing for various reasons. For example, device errors and problems during data collection, or patients not following instructions. This issue is particularly common in long-term monitoring, like Continuous Glucose Monitoring or Electronic Health Records[17].
- Privacy and Ethical Constraints: Regulations like HIPAA and GDPR place strict limits on how patient data can be used or shared. As a result, obtaining enough data to proper training and testing models becomes more challenge [18].

These challenges highlight the need for new methods that create effective models while still protecting privacy and including diverse data sources.

D. Synthetic Data in Medical AI

To address these challenges, researchers have started exploring synthetic data generation as a potential solution. Unlike real patients' records, synthetic data are artificially created. However, it is designed to reflect the patterns and structures of real patient data without showing any personal information[19]. This type of data can be applied at various stages of model development. For example, it can be used to training, testing, and comparing models. Synthetic data provides several advantages for diabetes care. It can help fix class imbalance, expand small datasets, model rare complications, and allow data sharing without breaking privacy rules[20]. Therefore, researchers can build stronger, more inclusive, and more generalizable AI models while remaining within the ethical and legal boundaries. Although synthetic data are not a new idea, their use in healthcare, especially in diabetes research, has only recently gained momentum.

III. SYNTHETIC DATA GENERATION TECHNIQUES

Over the past few years, synthetic data-generation methods have advanced significantly. These techniques offer

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several methods to create realistic artificial datasets. In diabetes research, choosing the right method depends on factors such as the type of data (tabular or time-series), how sensitive the data are, how complex it is, and how easily doctors can understand it. This section groups the main techniques into four categories: statistical, machine-learning-based, deep-learning-based, and hybrid or newer approaches.

A. Statistical Approaches

Statistical methods were among the first used to generate synthetic data. These techniques use mathematic-based models to understand patterns in real data, and then create new, similar data points.

Gaussian Mixture Models (GMMs): These model data are a mix of several Gaussian (bell-shaped) distributions. This is useful for creating new records from continuous multivariate data[21].

Bayesian Networks: Bayesian networks have emerged as a powerful technique for synthetic data generation, particularly in healthcare and medical research. These graphical models encode probabilistic relationships among variables, offering several advantages for data modeling and analysis[22]. Graph based models show the relationship between different features. They are useful when working with datasets that include multiple or combinations of data types.

Multivariate Normal Distributions: It plays an important role in synthetic data generation. It provides a strong foundation for producing datasets that are both realistic and statistically reliable [23]. These methods are particularly effective when dataset's features follow a normal distribution.

Copula-Based Models: Copulas are statistical tools that allow us to model individual variables separately from the relationships between them. This flexibility makes them suitable for handling both continuous and mixed-types. At the same time they preserve the dependencies between variables [24].

Statistical methods are especially useful for structured tabular diabetes data. This includes laboratory results, blood glucose measurements. Techniques like Gaussian Mixture Models or copula models can generate new patient records. These records closely reflect the patterns observed in the real-world data. These methods have several advantages. They are easy to interpret, simple to implement and required minimal computational resources. However, they also have limitations. For example, they may struggle to capture nonlinear relationships and complex interactions between features.

B. Machine Learning Based Approaches

ML based methods use predictive models to understand how real data behave. Once trained, these models can generate new synthetic samples. Compared to traditional statistical techniques, they are better at capturing complex relationships between features.

Decision Trees and Random Forests: Random Forests are powerful tools for classification and prediction. They can also be adapted for synthetic data generation. By modeling how features are distributed within decision paths. Random Forests can create realistic synthetic samples. In addition, they handle high dimensional data, missing values and unbalanced data sets very effectively. This combination of strengths makes them well suited for generating synthetic data by that closely resembles real world patterns [25].

The Synthetic Minority Oversampling Technique (SMOTE) is a popular method used in data science to handle unbalanced datasets. Sometimes, a dataset may have one class with many more examples than another. This imbalance can cause problems during analysis or model training. SMOTE is designed to fix the problem of class imbalance. It does this by generating new, artificial data points. Hence the dataset becomes more balanced. This can improve the performance of ML model. The method worked by looking at nearby samples in the feature space and blending them to create new points for underrepresented data. It is commonly used when there is class imbalance. This technique is flexible, with both conventional and advanced algorithms, including quantum computing.[26]

Adaptive Synthetic Sampling (ADASYN) is useful technique for addressing class imbalance in datasets. It can be particularly useful in diabetes prediction models. This method generates synthetic examples for the minority classes. It focuses on areas where the data are sparse or noisy. In this type of data learning of the model is minimal. ADASYN approach helps the model improve its performance in difficult regions of the dataset [27].

These methods work well for datasets like the Pima Indians Diabetes Dataset. In this dataset the number of diabetes cases is smaller than non-diabetes cases. They help balance the data by addressing this underrepresentation. Techniques such as SMOTE and ADASYN help fix this imbalance by creating new samples from the minority class. Therefore, these models are more likely to detect diabetes. These methods are easy to apply, scale efficiently, and integrate well with most ML workflows. However, these methods have certain limitations. If the synthetic samples do not reflect the real patterns and differences in the actual data, they may cause the model to overfit. This means the model might perform well on training data but poorly on new, unseen data.

C. Deep Learning Methods

Deep learning (DL) particularly generative models, has revolutionized synthetic data generation by learning complex high-dimensional patterns. These models are particularly effective for unstructured data such as images, time series, and sequences.

Generative Adversarial Networks (GANs): GANs use two types of neural network. These are generators and discriminators that compete to produce realistic synthetic data. Specialized versions such as CTGAN and MedGAN were designed for tabular and medical data[28].

Variational Autoencoders (VAEs): It compresses the data into a lower-dimensional space, reducing the number of features while keeping the most important information. Then reconstruct it to generate new synthetic samples. They are often used for continuous data. In biology, VAEs have been effectively used in omics data, including single-cell transcriptomics and metabolomics. For example, OntoVAE integrates ontologies into the latent space to produce interpretable pathways or phenotypic activities[29].

Recurrent GANs or LSTM-Based Generators: These designed for sequential data. These techniques have

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shown significant promise for generating synthetic time-series data, particularly for continuous glucose monitoring (CGM) devices. These models leverage the temporal dependencies inherent to sequential data to produce realistic synthetic records. GluGAN is a framework created to generate personalized glucose time series data. It uses Recurrent Neural Network (RNN) modules and combines both unsupervised and supervised learning methods to understand how glucose levels change over time in a latent space[30].

DL models such as GANs and VAEs have been effectively used to generate synthetic electronic health records. These models can use patient time-series data like insulin dosages. They can also include retinal images for diabetic retinopathy screening. They are capable of capturing complex, non-linear, multimodal patterns in how the disease progresses and how patient respond to treatment. However, these models have some challenges. They often require large datasets and high computing power. They can also be difficult to interpret. GANs are particularly tricky to train because they can be unstable. One common problem is mode collapse. This happens when the model generates only a limited variety of data.

D. Hybrid and Emerging Techniques.

Researchers have been looking for ways to overcome the limitations of individual techniques. One approach is to combine statistical methods with machine learning and deep learning. This combination can take advantage of the strengths of each. At the same time, new frameworks have been developed that are specifically aimed for medical applications. These frameworks address the unique challenges of working with healthcare data.

Bayesian GANs: Bayesian GANs extend standard GANs by modeling uncertainty. It helps them generate more robust and generalizable synthetic data. This makes them especially useful for handling limited or noisy datasets and for applications where understanding uncertainty is important. Compared with standard GANs, they often perform better in medical tasks. They are frequently used to create synthetic MRI, CT, and ECG data, which can help unknown patient information and simulate rare-disease cases[31].

Tabular GANs: CTGAN and TVAE models are specifically designed to generate synthetic data from tables that contain a mix of numerical and categorical variables. They can handle complex data distributions effectively. One example is CTGAN, which has shown promising results in producing high-quality synthetic data for electronic health records (EHR). It has also been used successfully to predict outcomes such as the length of hospital stay [32].

Federated Generative Models: These models can generate synthetic data from decentralized sources. They do this without requiring sensitive patient data to be centralized. It is especially useful when privacy is a concern. To achieve this, they use federated learning techniques, allowing distributed datasets to be trained without using the sensitive information [33].

Privacy-Preserving Synthetic Data: These techniques use approaches like differential privacy in generative models. It ensures synthetic data does not compromise patient privacy. For added security, researchers have developed federated GAN models. It combine homomorphic encryption with differential privacy. This allows multiple organizations to collaboratively train generative models without sharing raw patient data. Differential privacy enhanced GANs have been applied to real-world smart health datasets, like Fitbit data. They can generate realistic synthetic samples while preserving important patterns and protecting sensitive health information.[34].

Hybrid models have been used to generate synthetic profiles for diabetes patients. It can be used in decentralized healthcare systems. They can also simulate rare complications like diabetic ketoacidosis. In addition, these models can help improve the quality of data collected from wearable devices while protecting patient privacy. These models are highly flexible and can be adapted to different types of medical data. This makes them useful for datasets with complex structures and strict privacy requirements. However, implementing these methods is often challenging. Building and using hybrid models need expertise in data science, healthcare, and privacy regulations.

IV. APPLICATIONS IN DIABETES

Synthetic data has become an important tool in diabetes research. It helps overcome common challenges with real world data including privacy concerns, limited availability, class imbalances and inconsistent information. Researchers are using synthetic data across several key areas like electronic health records, time-series data like continuous glucose monitoring, medical imaging, decision support systems, and data augmentation for training models. In all of these areas, synthetic data supports innovation. It improves scalability, and enhances the performance of both machine learning models and clinical decision-making.

A. Electronic Health Records Synthesis

Electronic Health Records contained detailed long-term patient data. These included diagnoses, medications, laboratory results, and demographic information. EHRs are main resources for diabetes-related research and analysis. However, strict privacy laws like HIPAA and GDPR limit data sharing. This can make it difficult to reproduce research or validate models externally.

Generating synthetic EHR data has become a popular way to handle this challenge. It enables secure data sharing and supports early-stage model development. For example, platforms like Synthea can simulate synthetic EHRs that reflect the progression of both type 1 and type 2 diabetes. These simulations can include comorbidities and treatment pathways. Importantly, these datasets do not contain any real patient information. Research has shown that synthetic EHRs closely match the statistical and clinical characteristics of actual data. This makes them valuable for tasks such as epidemiological studies, selecting patient cohorts, and developing algorithms. At the same time, they help protect patient privacy.[6].

B. Time-Series and Continuous Glucose Monitoring (CGM)

CGM systems produce high-resolution time-series data. These data provide real-time insights into how glucose levels change throughout the day. They are especially important for studying hypoglycemic and hyperglycemic events. It adjust insulin doses and managing glucose levels in a personalized

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way. Collecting large CGM datasets, however, is not easy. Access to CGM devices can be limited. Patients may have difficulty following the monitoring rules consistently. Additionally, privacy regulations can make sharing and using these datasets more complicated [35].

Synthetic CGM data provides a way to overcome the limitations of real-world datasets. It can simulate realistic glucose trends. It can also mimic the effects of meals, insulin responses, and daily rhythms. This type of data helps researchers develop forecasting models. It also supports systems for detecting anomalies in glucose levels. For example, synthetic CGM datasets have been used to train deep learning models, such as LSTMs and Transformers. These models can predict glucose fluctuations and identify unusual patterns. By adding temporal variety and including rare events, synthetic data improves model performance. It also helps models generalize better, especially when real data are limited or unbalanced [36]

C. Medical Imaging

Medical imaging plays a major role in diagnosing complications related to diabetes. Some common imaging techniques include retinal fundus photography, foot thermography, and skin lesion imaging. These methods help detect conditions such as diabetic retinopathy, neuropathy, and foot ulcers. Despite their importance, datasets for these conditions often face challenges. The datasets are usually small. They can also be imbalanced, with some conditions underrepresented. In addition, labeling errors are common, which can reduce the accuracy of diagnostic models.

Researchers have tackled these challenges by using synthetic image-generation techniques, especially those based on GANs. For instance, conditional GANs can create synthetic retinal images, which help balance class distributions in diabetic retinopathy detection and improve model performance [37]. In a similar way, synthetic images of foot ulcers have been used for tasks like wound segmentation and severity classification, particularly when real data are hard to come by. These synthetic datasets not only make visual models more robust but also allow deep learning models, like CNNs, to be trained without putting patient privacy at risk. In a similar way, synthetic images of foot ulcers have been used for tasks like wound segmentation and severity classification, particularly when real data are hard to come by. These synthetic datasets not only make visual models more robust but also allow deep learning models, like CNNs, to be trained without putting patient privacy at risk.

D. Decision Support and Risk Prediction Models

Clinical decision support systems and risk prediction models are important tools for providing personalized care to people with diabetes. They can help guide decisions about lifestyle changes, medications, and the likelihood of developing complications. To be effective, these models need access to large, high-quality datasets that accurately reflect the patient population. In reality, though, such data is often limited or incomplete, making it harder for these tools to perform as intended.

Synthetic data provides a way to create complete patient profiles. These profiles can include demographic details, lifestyle habits, comorbidities, and laboratory results. Such data is useful for developing and testing risk prediction tools and clinical decision support systems in diabetes care. For example, synthetic data has been used to train ensemble classifiers that predict the onset of diabetes or its complications. These models can perform just as well as those trained on real patient data. Synthetic datasets also allow testing the system under rare or diverse conditions that are hard to find in actual clinical records [38].

E. Data Augumentation for Model Training

ML models in diabetes research often face several challenges. These include data imbalance, overfitting, and poor representation of rare events. To overcome these problems, researchers often use synthetic data generation. This approach helps augment existing datasets. It can be applied to different types of data, such as tabular data, time series data, and images.

In diabetes classification tasks, like detecting type 2 diabetes with the Pima Indians Diabetes Dataset, synthetic oversampling methods have proven useful. Methods such as ADASYN, SMOTE, and GAN-based techniques help balance class distributions. This, in turn, improves model performance. Synthetic data can also introduce controlled variations. These variations expand the feature space, helping models generalize better and reducing bias.

Synthetic data is particularly helpful in settings where privacy is a concern. For example, in federated or privacy-preserving setups, the original data cannot be shared or centralized [39].

V. DATASETS AND BENCHMARKS

Creating and evaluating synthetic data generation methods in diabetes research requires high-quality datasets. These datasets are essential for designing new methods. They are also needed for training and validating synthetic-data techniques. In this section, we review commonly used datasets in diabetes research. This includes both real-world and synthetic datasets.

These datasets have been used for various tasks. Examples include prediction, diagnosis, monitoring, and disease management. Datasets can be grouped by their type. The main types are tabular data, time-series data, and imaging data. For each type, we discuss its relevance and scope. We also consider their limitations in the context of synthetic data generation and evaluation.

A. Pima Indians Diabetes Dataset

PIDD is one of the most widely used datasets for predicting diabetes. It is hosted by the UCI ML Repository. Researchers often rely on it because of its accessibility and well-structured data[40]. The dataset includes 768 records of female patients of Pima Indian heritage, all aged 21 or older. It has eight clinical features, such as glucose level, BMI, age, and blood pressure, along with a binary outcome showing whether or not the person has diabetes.

- Data Type: Tabular
- Use Cases: Supervised learning, binary classification, model benchmarking
- Limitations: Small size, limited demographic diversity, absence of temporal or imaging data

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- Relevance to Synthetic Data**:** Frequently used to generate synthetic tabular data to address class imbalance and overfitting[41] [42].

B. MIMIC-III and MIMIC-IV

The Medical Information Mart for Intensive Care (MIMIC) database, maintained by the MIT Lab for Computational Physiology, is an extensive critical care dataset. MIMIC-III includes data from over 60,000 ICU admissions, covering lab results, vital signs, and medications. MIMIC-IV builds on this by providing more detailed information and a better-organized, modular structure.

- Data Type: Time-series, tabular, clinical text
- Use Cases: Prognostic modeling, disease progression analysis, treatment recommendation
- Limitations: Requires data usage certification, complex pre-processing, and limited diabetes-specific labels.
- Relevance to Synthetic Data: Used to simulate patient trajectories, generate longitudinal synthetic data, and evaluate privacy-preserving synthetic techniques[43] [44].

C. National Health and Nutrition Examination Survey (NHANES)

NHANES is a extensive program carried out by the CDC, combining interviews and physical examinations. It contains extensive health and lifestyle data relevant to metabolic disorders including diabetes.

- Data Type: Tabular, survey data
- Use Cases: Epidemiological studies, public health surveillance, feature selection
- Limitations: Complex sampling design; missing values; non-continuous monitoring
- Relevance to Synthetic Data**:** Generated synthetic demographic and lifestyle variables were applied in fairness testing and generalizability analysis[45].

D. OhioT1DM and OhioT2DM Datasets

Published by the UCI repository, the Ohio Type 1 and Type 2 Diabetes datasets comprise anonymized continuous glucose monitoring (CGM) data from real-world patients.

- Data Type: Time-series (CGM, insulin, meal events)
- Use Cases: Blood glucose prediction, hypoglycemia detection, real-time modeling
- Limitations: Small sample size (~12 patients); inter-patient variability
- Relevance to Synthetic Data**:** Frequently used to train generative models for synthetic time series and simulate insulin-glucose dynamics [46] [47].

E. EyePACS and Messidor

These datasets are important in the field of diabetic retinopathy detection and grading. EyePACS, used in the Kaggle DR challenge, consists of over 88,000 retinal images, whereas Messidor includes 1,200 images with expert annotations.

- Data Type: Medical imaging (fundus photographs)
- Use Cases: Image classification, segmentation, lesion detection
- Limitations: Varying image quality, label imbalance, domain shift
- Relevance to Synthetic Data**:** Serve as the ground truth for training GAN-based image generators used in data augmentation and domain adaptation studies[48].

F. UCI diabetes dataset

This dataset includes data from a diabetes patient readmission study in 130 U.S. hospitals over 10 years, comprising over 100,000 records with more than 50 attributes related to medical history and hospitalization.

- Data Type: Tabular, hospital records
- Use Cases**:** Readmission prediction, treatment recommendation
- Limitations: High dimensionality, missing codes, class imbalance
- Relevance to Synthetic Data: Suitable for studying generative methods in structured EHR settings[49].

G. Synthea™

Synthea is an open-source tool that generates synthetic patient data. It simulates how diseases progress and how treatments are given under different conditions, including diabetes.This produces realistic, privacy-preserving, and EHR-like data.

- Data Type: Tabular, longitudinal, synthetic EHR
- Use Cases: Benchmarking ML pipelines, system simulations, pretraining
- Limitations: Modeled, not real; potential for unrealistic rare event distribution
- Relevance to Synthetic Data: Directly generates synthetic diabetes cohorts with customizable demographics and progression models [50] [51].

High-quality datasets are essential for modern diabetes research, because they support both traditional and synthetic data approaches. Real-world datasets offer valuable insights but often suffer from privacy concerns, missing values, and class imbalance. Despite these challenges, they remain critical for training and validating synthetic data-generators. Synthetic data generators, such as Synthea, help overcome these limitations by producing scalable, diverse, and privacy-compliant datasets. To ensure reproducibility and comparability in synthetic data research on diabetes, it is essential to create consistent benchmarks and assessment frameworks.

VI. EVALUATION METHODS

Evaluating synthetic data is very important to make sure it works well, especially in diabetes research. Synthetic data should reflect the key features of real-world data. At the same time, it needs to overcome the challenges involved in data generation. Researchers use different methods to evaluate synthetic data. These methods look at several aspects. They consider the quality of the data and how well models perform using it. Privacy and security are also important factors. Finally, evaluation checks whether the data can be applied to real-world diabetes research and clinical practice.

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A. Data Quality Assessment

The quality of synthetic data is largely measured by how closely it resembles real-world data. This is important because synthetic data can only be useful for machine learning and clinical decision-making if it preserves the same distributions and statistical characteristics as the original data. To ensure this, researchers have developed several methods to assess the quality of synthetic data.

Statistical Comparisons: Basic statistical methods are often used to evaluate synthetic data. Researchers compare means, variances, and higher-order moments. For example, skewness and kurtosis can show how closely synthetic data replicate real-world data. These comparisons help ensure that the synthetic data maintain distributional properties similar to the original dataset. This is particularly important when working with complex medical variables in diabetes research[52].

Visual Inspection: Dimensionality reduction techniques, such as t-distributed Stochastic Neighbor Embedding (t-SNE) and Principal Component Analysis (PCA), are often used to visualize synthetic data. These methods map data into lower-dimensional spaces. By projecting data into two or three dimensions, researchers can compare the distribution, spread, and clustering patterns of synthetic and real data. In diabetes research, such visualizations help show the relationships between key variables, including glucose levels, insulin dosage, and patient demographics [53].

Advanced Distance Measures: Advanced metrics are increasingly used to evaluate how closely synthetic data match real data. Examples include the Wasserstein Distance (also called Earth Mover's Distance) and the Fréchet Inception Distance (FID). The Wasserstein Distance measures the minimal effort needed to transform one distribution into another. FID is mainly used for images. It compares the feature distributions of real and synthetic images. These metrics have been applied in GAN models to generate high-quality synthetic retinal images. For instance, a recent study used a Wasserstein GAN with gradient penalty (WGAN-GP) to produce realistic images of diabetic retinopathy [37]. These measures are particularly valuable for assessing the quality of synthetic medical images, including retinal scans and X-rays, in diabetes research.

The goal of data quality assessment is to ensure that synthetic data accurately reflect the key characteristics of real data. This includes variables such as glucose levels, patient history, and other clinical measurements. Capturing these details is essential for robust diabetes research and for building reliable predictive models.

B. Model Performance Evaluation

After generating synthetic data, it is important to evaluate how useful it is for training machine learning models. Synthetic data can be used to augment real datasets. They can also serve as the main training data when real data are limited. Researchers commonly use several evaluation metrics to measure how well synthetic data improve model performance.

Accuracy, Precision, Recall, and F1-Score: Classical evaluation metrics are used to assess the performance of models trained with synthetic data. In diabetes applications, accuracy shows how well a model can correctly classify patients with and without diabetes. Precision and recall, on the other hand, focus on detecting true positives. They measure how effectively the model identifies diabetic patients and avoids misclassifying them as nondiabetic [54].

ROC-AUC: The receiver operating characteristic curve and its area under the curve are important metrics for evaluating classifier performance. In diabetes prediction models, the ROC-AUC measures the trade-off between sensitivity (recall) and specificity. This is especially important in medical contexts, where both false positives and false negatives can have serious consequences for patient care [55].

Model Robustness and Generalizability: Beyond traditional metrics, it is important to check how well a model trained on synthetic data performs on real, unseen data. This can be done using cross-validation. In this approach, the model is tested on a hold-out set or an external dataset, such as data from a different hospital or patient population. This helps evaluate the model's robustness and its ability to make accurate predictions across diverse patient groups[56].

Model performance evaluation also extends to the assessment of fairness in diabetes prediction models, ensuring that the models do not introduce bias towards specific demographic groups (e.g., age, gender, ethnicity) in synthetic datasets. Techniques such as fairness-aware learning and disparate impact analysis can be applied to examine the potential biases in synthetic data.

C. Privacy and Security Evaluation

Healthcare data is highly confidential. Because of this, privacy-preserving synthetic data is very important. It allows researchers to work with data while keeping patient information secure. Several methods have been developed to evaluate the privacy and security of synthetic datasets.

Re-identification Risk: A major concern with synthetic data is the risk that it could be reverse-engineered. This might allow someone to identify individuals in the original dataset. To address this, techniques such as k-anonymity and l-diversity are used. These methods help assess whether synthetic data could be linked back to real patient identities[19]. Diabetes-related apps often use personal health information. This can include blood sugar levels and insulin usage. It is very important to ensure that individuals cannot be identified from this data.

Differential Privacy: This approach involves adding noise to the data to protect personal privacy. At the same time, it allows the data to be used for analysis. Differential privacy techniques are designed to safeguard individual information when sharing synthetic data. They ensure that no one can identify a specific person, even if they have access to additional related information [57].

Adversarial Testing: Adversarial techniques are used to check if synthetic data can be exploited to reveal sensitive information. In diabetes research, this might involve trying to identify patients or their specific health conditions from synthetic data used in predictive models [58].

It is very important to ensure that synthetic data is private and secure. Doing so helps make sure the data complies with laws such as the Health Insurance Portability and Accountability Act. These regulations protect medical information when it is used for research.

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D. Real-world applicability and domain specific evaluation

A key measure of synthetic data's usefulness is how well it supports real-world diabetes applications. This involves evaluating whether synthetic data can help address practical challenges, such as limited data availability, improving model training, and supporting clinical decision-making.

Clinical Decision-Making: Synthetic data can help improve decision-making by providing more complete datasets for training predictive models. For example, they can be used to augment real EHR data. This ensures that decision support systems have diverse and comprehensive information to help healthcare providers diagnose and manage diabetes effectively.

Underrepresented Populations: Synthetic data can help improve the representation of underrepresented populations in diabetes research. This includes minority groups and patients with rare forms of diabetes [59]. Researchers can generate synthetic samples from these populations. Using these samples, they can develop models that reflect a broader range of patient characteristics.

Data Augmentation in Model Training**:** In diabetes-related tasks, such as analyzing retinal images, synthetic data can be used to create additional training samples. This helps improve model accuracy, robustness, and generalization. When real-world data are limited or costly to collect, synthetic data provides a practical alternative. It allows machine learning models to be trained without relying on extensive real data

Challenges and limitations

Synthetic data have great potential for diabetes-related applications. However, there are several important challenges and limitations that must be addressed. Tackling these issues is essential for using synthetic data ethically and effectively in both clinical and research settings. This section examines the main challenges in detail. These include concerns about data quality, model performance, privacy, regulatory compliance, scalability and real-world applicability.

E. Data quality and authenticity

One major challenge in creating synthetic data is making sure it accurately reflects real-world data. Diabetes data, such as glucose levels, patient demographics, and clinical conditions, are often complex and highly variable. This variability makes it difficult to capture all the detailed and interconnected factors involved in the disease. For example, glucose levels are influenced by many factors, including diet, medication, and stress, which vary from person to person. Synthetic data may not fully capture these individual differences and complexities. As a result, it may be insufficient to represent the full diversity found in real-world populations with diabetes.

Another important concern is the potential loss of authenticity when generating synthetic data. These datasets may fail to capture subtle patterns, such as correlations and causal relationships, that exist in real-world data [60]. Missing details in synthetic data can affect how well machine learning models perform. This is especially important for diabetes-related applications. Accurate prediction and diagnosis rely on models that generalize effectively to real-world situations. If synthetic data is not authentic, it can reduce the reliability and effectiveness of these models in practice.

Model overfitting and Bias

Overfitting is an important concern when using synthetic data in machine learning models. It occurs when a model performs very well on synthetic data but fails to generalize to new, unseen real-world data. This problem often arises if synthetic data are too simplistic or lack diversity. In such cases, the model may simply memorize patterns in the synthetic dataset. It does not truly learn the underlying data distributions [61].

In addition, synthetic data can introduce biases. This happens when the generative process does not capture the full range of variability in diabetes-related data [47]. For example, synthetic datasets may underrepresent certain demographic groups. This can lead to biased models that perform poorly for those populations. This issue is especially important in diabetes research, where healthcare outcomes often vary across ethnicities, socioeconomic statuses, and age groups. Ensuring that synthetic data reflect these variations is crucial. Doing so helps prevent the perpetuation of biases and supports equitable healthcare outcomes.

Privacy concerns and Security

One of the main reasons for using synthetic data in healthcare is to protect patient privacy. However, despite their advantages, synthetic datasets still pose privacy and security challenges. Even though they are not directly derived from individual patient records, there is still a risk of re-identification. This risk is higher in smaller datasets or those containing highly detailed features. It becomes even more significant if the generative models used to create synthetic data are not robust enough to block all possible re-identification pathways [6].

Another important challenge is balancing data utility with patient privacy. Synthetic data should remain useful while protecting sensitive information. Techniques like differential privacy are often applied to enhance privacy. However, there is still uncertainty about how effective these protections are, especially in healthcare settings. This issue is particularly critical in sensitive areas such as diabetes management. Finding the right balance between data utility and privacy continues to be a major challenge [62].

F. Regulatory and ethical issues

The use of synthetic data in healthcare faces regulatory and ethical challenges. Many governing bodies, such as the U.S. Food and Drug Administration (FDA), India's Central Drugs Standard Control Organization (CDSCO), and the Indian Council of Medical Research (ICMR), have not yet set clear guidelines for using synthetic data in clinical applications. This lack of standardization can slow the adoption of synthetic data in research and clinical practice. In particular, synthetic datasets may not meet the strict requirements needed for clinical trials, where patient data must accurately represent real-world conditions and outcomes.

Creating synthetic clinical data using advanced models like GANs and VAEs raises ethical concerns. One issue is transparency, as these models often lack interpretability. This makes it hard to trace or validate the

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generated outputs. If not carefully designed, synthetic data can hide healthcare disparities or even reinforce biases. Ensuring fairness, explainability, and thorough validation is essential to maintain trust. Regulatory oversight and ethical frameworks from organizations such as the FDA, CDSCO, and ICMR are crucial for responsible use.

G. Scalability and computational complexity

Another major limitation of synthetic data generation is its scalability. Working with large, high-dimensional datasets can be computationally complex and resource-intensive[28]. For example, medical imaging data or longitudinal time-series data from continuous glucose monitoring (CGM) devices are often high-dimensional. Generating such data effectively requires significant computational resources. Advances in deep learning techniques, such as GANs and VAEs, have shown promise for creating synthetic data from complex datasets. However, these methods still demand substantial computational power, especially when generating data at large scale.

Furthermore, the scalability of synthetic data generation methods needs careful evaluation in real-time clinical settings. In these environments, quick data processing is essential. High computational costs and resource demands for producing accurate synthetic data can limit their adoption, especially in resource-limited settings. Addressing these challenges is crucial to support the widespread use of synthetic data across diverse healthcare contexts.

H. Real world applicability

Synthetic data generation methods have been tested in research settings. However, their effectiveness in real-world scenarios is still uncertain. Diabetes-related datasets in clinical settings are influenced by many factors. These include diverse patient populations, varying healthcare conditions, and regional differences in medical practices. Synthetic data created in controlled research environments may not capture all of these real-world variations. As a result, applying them reliably in different clinical settings can be challenging.

Furthermore, the ability of models trained on synthetic data to generalize to real-world scenarios is a major concern. Models may struggle in actual clinical settings if the synthetic data does not capture the full range of conditions seen in practice. Ensuring that synthetic data can be applied effectively across diverse clinical environments is critical for their success.

VII. Future directions

The use of synthetic data in diabetes applications has shown great potential. However, many exciting opportunities are still unexplored. As generative models, machine-learning algorithms, and healthcare technologies evolve, synthetic data generation has the potential to transform diabetes research, care, and management. This section highlights the key areas where progress in this field is expected.

A. Advances in generative models

Generative models, including GANs, VAEs, and diffusion models, have made significant progress in generating synthetic data. These models are being applied in many sectors, including healthcare. The future of synthetic data in diabetes applications is expected to benefit greatly from further advancements in these models. GANs, in particular, are likely to improve the generation of high-quality datasets. These datasets can closely resemble real data and remain clinically relevant. GANs can capture complex patterns in diabetes data. This includes glucose variations, patient demographics, and comorbidities. Capturing these patterns can enhance predictive model development. In addition, emerging architectures like conditional GANs and StyleGANs may offer better control over data generation. They could enable the creation of synthetic data tailored to specific subgroups. This includes rare conditions or underrepresented populations [63] [64].

VAEs are effective at capturing complex patterns in data. This ability can be used to generate synthetic diabetes datasets that more closely reflect real-world data. Such datasets can help address common issues in diabetes research, such as missing or imbalanced data.

Diffusion models, which have been successful in generating high-resolution images, can also create synthetic data in areas such as CGM and medical imaging. These models can produce high-quality datasets. Using such data can improve model training. It can also enhance the diagnostic and prognostic capabilities in diabetes care [65].

B. Personalized Synthetic data

Personalized synthetic data is an important step toward creating accurate medical datasets. It involves generating data tailored to individual patients or specific groups in diabetes care. This approach can greatly improve predictive models and support better clinical decisions. Current methods often produce general datasets. These general datasets may miss important details, such as variations in disease progression, lifestyle, or genetic factors.

Personalized synthetic data can help fill this gap by generating datasets that reflect the specific health profiles of individual patients, including those with rare forms of diabetes. This allows clinicians to test models and treatments using data that more accurately represents their patients. As a result, care can be more targeted and effective. Personalized synthetic data can also improve the accuracy and reliability of machine learning models used for diabetes prediction, management, and treatment planning[66].

C. Cross domain data generation

In diabetes care, combining different types of medical data is essential for building well-rounded healthcare solutions. In the future, synthetic data may extend beyond single sources, such as EHRs or glucose readings. It could integrate information from multiple areas, including continuous glucose monitoring (CGM), medical imaging, genetics, and lifestyle habits. This cross-domain approach would help create more complete and accurate models for diabetes care.

Generating synthetic data across multiple domains can provide a deeper and more complete understanding of patient health. This approach can lead to more precise and personalized predictive models. For example, combining EHR and CGM data can create synthetic profiles that capture both medical history and real-time glucose patterns. Adding medical images, such as retinal scans or X-rays, along with genetic data, can further enhance diagnosis, treatment planning, and outcome prediction in diabetes care.

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Cross domain synthetic data generation can greatly improve the training of machine learning models. Combining different types of data provides a more complete picture of the factors contributing to diabetes and its complications. These complications include heart disease, diabetic retinopathy, and kidney damage. A more comprehensive dataset helps build more reliable and accurate models for prediction and treatment.

D. Ethical and regulatory advancements

The increasing use of synthetic data in healthcare requires strong ethical and regulatory frameworks. These frameworks should protect privacy, ensure transparency, and support accountability. Because synthetic data can resemble real patient information, there is a risk of re-identification. This risk is especially high in small or highly detailed datasets. Future research should focus on developing reliable methods that protect privacy while keeping data useful for training and testing models.

Regulatory bodies must set clear guidelines for using synthetic data in clinical settings. This is especially important in diabetes care. It is essential that these datasets meet ethical standards and comply with regulatory rules. Doing so will help them gain wider acceptance in healthcare. As the use of synthetic data grows, there is a greater need for standard methods to verify their quality, safety, and usefulness. These measures will ensure that synthetic datasets are reliable and suitable for clinical use.

E. Integration into clinical workflows

Synthetic data holds great promise for diabetes research. However, a major challenge is integrating it into real-world clinical workflows. For synthetic data to make a real impact, it must go beyond research and fit seamlessly into daily clinical practice. This ensures compatibility with existing clinical systems and allows it to be used effectively as a decision support tool. Overcoming these challenges is essential to unlocking its full potential in patient care.

Future advancements should focus on creating user-friendly systems that allow clinicians to access and use synthetic data easily in decision making. Synthetic data should also enhance the clarity and usefulness of AI models in clinical settings. For example, integrating synthetic data into electronic health records (EHRs) or clinical support tools can let healthcare providers test and fine-tune models without risking patient privacy. This approach supports the safer and more effective adoption of AI in diabetes care.

F. AI and ML driven model training with synthetic data

The use of synthetic data for training machine learning models is expected to grow, especially for addressing gaps in data for rare or underrepresented conditions. In diabetes research, available data are often limited for certain groups. These include children with type 1 diabetes or older adults with multiple health issues. Synthetic data can help fill these gaps by generating large and diverse datasets that include these populations. This, in turn, supports the development of more accurate and generalizable models for diabetes care.

In the future, synthetic data could improve the robustness and fairness of machine-learning models. By generating diverse datasets that better capture variations in patient populations, models can be trained to reduce bias. This allows for more equitable predictions across different demographic groups. Moreover, synthetic data can help test models in rare scenarios, such as predicting diabetic retinopathy or hypoglycemia. This approach would enhance the clinical applicability of AI systems in diabetes care.

VIII. CONCLUSION

This survey highlights the potential of synthetic data generation in diabetes research. Synthetic data offer several benefits. They can improve access to data, enhance model training, and help address privacy concerns in healthcare. Synthetic data can also overcome limitations of real-world data, especially when patient information is scarce, imbalanced, or sensitive. However, challenges remain in ensuring the quality and authenticity of these datasets. Issues such as model overfitting, biases, privacy risks, and integration into clinical workflows must be addressed. Future advances in generative models, such as GANs, VAEs, and diffusion models, are expected to produce more accurate and diverse datasets. Personalized synthetic data tailored to specific patient populations could further improve predictive models. Combining different types of data, such as electronic health records, continuous glucose monitoring, and medical imaging, can also enhance diabetes care. The success of synthetic data depends on developing ethical and regulatory frameworks that ensure privacy, transparency, and accountability. These frameworks will support the safe integration of synthetic data into clinical practice. Although challenges remain, ongoing research and technological progress are likely to unlock the full potential of synthetic data. This will help advance personalized medicine and improve patient outcomes in diabetes care.

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